

Iterative compression

Iterative compression is an approach which answers the question if there exists a solution of the size at most k . We start with only a part of the original instance (e. g. induced subgraph) and try to find a correct solution. Then, we extend this instance by an element from an original instance and now we have a solution of size $k+1$. Using a compression algorithm we try to reduce this to k . We can successively repeat the steps described above until we have the answer to the original question.

1. Implement Minimum Vertex Cover using Iterative compression (check if for a graph, there exists a cover of size at most k).